

Assignment 5 LA MCQ

Each complete question and its options are followed by the correct answer.

Question 1: Which of the following represents a vector? A. 25 B. $[2, 5, 7]$ C. $2 + 5 + 7$ D. $x = 10$

Answer: B. $[2, 5, 7]$

Question 2: If $x = [2, 3]$ and $y = [4, 5]$, what is $x + y$? A. $[6, 8]$ B. $[8, 15]$ C. $[2, 2]$ D. $[4, 3]$

Answer: A. $[6, 8]$

Question 3: If $x = [2, 3]$, what is $4x$? A. $[6, 7]$ B. $[8, 12]$ C. $[2, 12]$ D. $[4, 3]$

Answer: B. $[8, 12]$

Question 4: What does the magnitude of a vector represent? A. Its direction B. Its length C. Its number of elements D. Its number of dimensions only

Answer: B. Its length

Question 5: What is the Euclidean norm of the vector $[3, 4]$? A. 3 B. 4 C. 5 D. 7

Answer: C. 5

Question 6: Two vectors are orthogonal when: A. They have the same magnitude B. Their dot product is zero C. Their values are identical D. Their magnitudes are zero

Answer: B. Their dot product is zero

Question 7: What is the dot product of $[1, 2]$ and $[3, 4]$? A. 5 B. 7 C. 10 D. 11

Answer: D. 11

Question 8: If two non-zero vectors have a dot product of zero, they are: A. Parallel B. Orthogonal C. Identical D. Linearly dependent

Answer: B. Orthogonal

Question 9: A matrix with 4 rows and 3 columns has dimensions: A. 4×3 B. 3×4 C. 12×1 D. 4×4

Answer: A. 4×3

Question 10: Which of the following is a square matrix? A. 2×3 B. 3×2 C. 4×4 D. 5×3

Answer: C. 4×4

Question 11: What does transposing a matrix do? A. Multiplies all elements by -1 B. Interchanges rows and columns C. Converts all elements to zero D. Divides every element by its mean

Answer: B. Interchanges rows and columns

Question 12: If matrix A has dimensions 3×5 , what are the dimensions of A^T ? A. 3×5 B. 5×3 C. 3×3 D. 5×5

Answer: B. 5×3

Question 13: Which pair of matrices can be added? A. 2×3 and 3×2 B. 2×3 and 2×3 C. 3×4 and 4×3 D. 2×2 and 3×3

Answer: B. 2×3 and 2×3

Question 14: Matrix multiplication AB is possible when: A. Rows of A = rows of B B. Columns of A = columns of B C. Columns of A = rows of B D. Both matrices are square

Answer: C. Columns of A = rows of B

Question 15: If A is 3×4 and B is 4×2 , what is the dimension of AB ? A. 3×2 B. 4×4 C. 2×3 D. 3×4

Answer: A. 3×2

Question 16: If A is 2×3 and B is 3×4 , what is the dimension of AB? A. 2×4 B. 3×3 C. 4×2 D. 2×3

Answer: A. 2×4

Question 17: Which statement about matrix multiplication is generally true? A. $AB = BA$ always B. AB and BA are always defined C. AB is not necessarily equal to BA D. Matrix multiplication is only possible for square matrices

Answer: C. AB is not necessarily equal to BA

Question 18: A linear combination of vectors involves: A. Only multiplication B. Only addition C. Scaling vectors and adding them together D. Multiplying matrices only

Answer: C. Scaling vectors and adding them together

Question 19: Suppose: $v_1 = [1, 0]$ $v_2 = [0, 1]$ Which vector can be represented as a linear combination of v_1 and v_2 ? A. $[2, 3]$ B. $[1, 1]$ C. $[5, -2]$ D. All of the above

Answer: D. All of the above

Question 20: If: $v_2 = 3v_1$ then v_1 and v_2 are: A. Orthogonal B. Linearly dependent C. Linearly independent D. Perpendicular

Answer: B. Linearly dependent

Question 21: A set of vectors is linearly independent if: A. Every vector has the same magnitude B. No vector can be represented as a combination of the others C. All vectors are perpendicular D. All vectors contain positive values

Answer: B. No vector can be represented as a combination of the others

Question 22: If one feature can be exactly calculated from another feature, the two features are: A. Orthogonal B. Independent C. Redundant D. Random

Answer: C. Redundant

Question 23: The rank of a matrix represents: A. Total number of elements B. Number of rows C. Number of linearly independent rows or columns D. Number of diagonal elements

Answer: C. Number of linearly independent rows or columns

Question 24: A matrix whose columns are all multiples of one another will have: A. Rank 0 B. Rank 1 C. Rank equal to the number of columns D. Infinite rank

Answer: B. Rank 1

Question 25: If a dataset has 10 features but some features are completely redundant, the effective number of independent directions may be: A. Greater than 10 B. Exactly 10 C. Less than 10 D. Infinite

Answer: C. Less than 10

Question 26: A vector can be interpreted geometrically as: A. A point only B. A direction and magnitude C. A scalar value only D. A matrix

Answer: B. A direction and magnitude

Question 27: What happens when a vector is multiplied by a scalar greater than 1? A. Its direction always changes B. Its magnitude increases C. It becomes orthogonal D. It becomes zero

Answer: B. Its magnitude increases

Question 28: Multiplying a vector by -1: A. Changes its magnitude to zero B. Reverses its direction C. Makes it orthogonal to every vector D. Removes its dimensions

Answer: B. Reverses its direction

Question 29: The angle between two vectors can be determined using: A. Their dot product and magnitudes B. Their number of elements only C. Their matrix dimensions D. Their rank only

Answer: A. Their dot product and magnitudes

Question 30: If two vectors point in exactly the same direction, the angle between them is: A. 0° B. 45° C. 90° D. 180°

Answer: A. 0°

Question 31: The projection of one vector onto another can be understood as: A. The component of one vector along the direction of another B. The difference between two vectors C. The magnitude of both vectors multiplied together D. A completely new random vector

Answer: A. The component of one vector along the direction of another

Question 32: If vector a is projected onto vector b , the projection tells us: A. How much of a lies in the direction of b B. How many elements are in b C. The rank of a D. The determinant of b

Answer: A. How much of a lies in the direction of b

Question 33: If two vectors are orthogonal, the projection of one onto the other is: A. Maximum B. Zero C. Equal to its magnitude D. Always one

Answer: B. Zero

Question 34: Why is projection useful when representing data? A. It can represent information using a chosen direction B. It always increases dimensionality C. It removes all information D. It changes categorical data into numerical data

Answer: A. It can represent information using a chosen direction

Question 35: An eigenvector of a matrix is a vector whose: A. Direction remains unchanged after the transformation, apart from scaling B. Magnitude always becomes zero C. Values always become positive D. Number of elements changes

Answer: A. Direction remains unchanged after the transformation, apart from scaling

Question 36: In the equation: $A v = \lambda v$ what is v ? A. Matrix B. Eigenvalue C. Eigenvector D. Determinant

Answer: C. Eigenvector

Question 37: In the equation: $A v = \lambda v$ what is λ ? A. Eigenvector B. Eigenvalue C. Matrix rank D. Matrix inverse

Answer: B. Eigenvalue

Question 38: If an eigenvalue is 5, what does this generally indicate about its associated eigenvector? A. The vector is rotated by exactly 5° B. The vector is scaled by a factor of 5 C. The vector becomes zero D. The vector becomes orthogonal

Answer: B. The vector is scaled by a factor of 5

Question 39: If an eigenvalue is zero, then: A. The associated transformed vector is zero B. The eigenvector does not exist C. The matrix must be identity D. The vector becomes larger

Answer: A. The associated transformed vector is zero

Question 40: Eigenvectors are useful for identifying: A. Important directions associated with a matrix transformation B. Missing values in a dataset C. Number of observations D. Categorical variables

Answer: A. Important directions associated with a matrix transformation

Question 41: Variance measures: A. The average value of a variable B. How much a variable varies around its mean C. The number of observations D. The relationship between two different variables

Answer: B. How much a variable varies around its mean

Question 42: If all values in a variable are exactly the same, its variance is: A. 0 B. 1 C. -1 D. Infinite

Answer: A. 0

Question 43: Covariance measures: A. How two variables vary together B. The mean of one variable C. The number of features D. The magnitude of a vector

Answer: A. How two variables vary together

Question 44: A positive covariance generally indicates: A. Both variables tend to increase together B. One variable always decreases C. There is no relationship D. Both variables are constant

Answer: B. One variable always decreases

Question 45: A negative covariance generally indicates: A. Both variables increase together B. One variable tends to increase when the other decreases C. Both variables are identical D. Both variables have zero variance

Answer: B. One variable tends to increase when the other decreases

Question 46: If two variables have covariance close to zero, it indicates: A. They definitely have no relationship of any kind B. There is little or no linear relationship detected C. They must be identical D. They must be orthogonal vectors

Answer: B. There is little or no linear relationship detected

Question 47: Consider a dataset represented as a matrix where rows represent observations and columns represent features. What does each row represent? A. A feature B. An observation/sample C. A matrix transformation D. An eigenvector

Answer: B. An observation/sample

Question 48: In a dataset matrix, each column generally represents: A. An observation B. A feature/variable C. An eigenvalue D. A class boundary

Answer: B. A feature/variable

Question 49: Suppose two features contain almost exactly the same information. What is the most likely consequence? A. Increased redundancy B. Increased independence C. Guaranteed increase in useful information D. Orthogonality

Answer: A. Increased redundancy

Question 50: A dataset contains 100 features, but many of them are strongly correlated. What might this suggest mathematically? A. The data may have fewer effective independent directions than 100 B. The data must have exactly 100 independent directions C. The data cannot be represented as a matrix D. All features are orthogonal

Answer: A. The data may have fewer effective independent directions than 100

Question 51: A transformation changes a vector from $[2, 1]$ to $[6, 3]$. What can you immediately conclude? A. The direction has changed B. The vector has been scaled by 3 C. The vector has become orthogonal D. The vector has been translated

Answer: B. The vector has been scaled by 3

Question 52: Suppose three vectors lie along the same line. Their maximum number of linearly independent directions is: A. 0 B. 1 C. 2 D. 3

Answer: B. 1

Question 53: If a matrix has highly dependent columns, its rank will generally be: A. Higher B. Lower C. Infinite D. Unaffected in all cases

Answer: B. Lower

Question 54: A vector has a large magnitude but points in the same direction as another vector. What can be said about the two vectors? A. They must be orthogonal B. They may be scalar multiples of each other C. They must be independent D. They must have zero dot product

Answer: B. They may be scalar multiples of each other

Question 55: Which mathematical concept is most directly related to finding important directions in a linear transformation? A. Eigenvectors B. Mean C. Median D. Mode

Answer: A. Eigenvectors